

Year 9

Mini Test 1



Baldivis
Secondary College

Number and Algebra

Name: Answers

Total Marks: 32 marks

Total Time: 30 minutes

Show working and answers on this sheet. Show working in sufficient detail to support your answers. Incorrect answers without supporting reasoning may be allocated zero marks.

Questions 1-8 are multiple choice one mark each.
Circle the correct answer.

1. 4^3 in expanded form is:

☒ A 64

☒ B

$4 \times 4 \times 4$

C

$3 \times 3 \times 3 \times 3$

D 4^3

E

12

2. $2^{12} \div 2^7$ is equal to:

A 4

B 8

C 16

D 25

☒ E 32

3. $2x^7 \times 7x^3$ is equivalent to:

A $14x^{21}$

☒ B $14x^{10}$

C $14x^4$

D $9x^{10}$

E $9x^{21}$

4. $-2x^0 + (9x)^0$ simplifies to:

A -2

☒ B -1

C 0

D 2

E 7

5. $\frac{6a^7b^5}{3a^2b^8}$ simplifies to:

☒ A $\frac{2a^5}{b^3}$

B $\frac{b^5}{2a^3}$

C $\frac{2a^3}{b^5}$

D $\frac{2b^3}{a^5}$

E $\frac{a^5}{2b^3}$

6. When we expand brackets, we use the distributive law. Which of the following does *not* represent the distributive law?

A $a(b + c) = ab + ac$

B $-a(b + c) = -ab - ac$

C $-a(b - c) = -ab - ac$

D $-a(b - c) = -ab + ac$

E $a(b - c) = ab - ac$

7. Which of the following expressions is equivalent to $2(5m + 1)$?

A $7m + 1$

B $10m + 1$

C $10m + 2$

D $12m$

E $11m$

8. Which of the following expressions is equivalent to $5c + 10$?

A $2c \times 3c + 10$

B $5c + c + 10$

C $15c$

D $c + 16 + 4c - 6$

E $2c + 16 - 3c - 6$

9. Evaluate

a. ~~2^4~~ ~~2^4~~ $3^3 = 27$

b. $7^0 = 1$

c. $5^{-2} = \frac{1}{25}$

d. ~~$(2y^0)^3$~~ 8

[4 marks]

10. Simplify the following expressions

a. $6 \times m \times m \times m$

$6m^3$

b. $3bc \times 2ab$

~~$6b^2$~~ $6ab^2c$

c. $2p - 3p^2 + 7p + 5p^2 - 3$

$9p + 2p^2 - 3$

d. $t^3 \times t^5$

t^8

e. $(n^3)^4$

n^{12}

f. $3a^4b^5 \times 7a^7b$

$21a^{11}b^6$

g. $\frac{3(e^4)^5 \times e^2}{9e^{-3}}$

$\frac{e^{25}}{3}$ (2 marks)

[8 marks]

11. Which two algebraic expressions are equivalent?(Circle your choices)

A $7x + 7$

☒ B $7x + 7y$

C $7xy$

D $x + 7y$

☒ E $7(x + y)$

[1 mark]

12. Expand the following and simplify when possible.

a. $5(x - 1)$

$5x - 5$ ✓

b. $2x(5 - x)$

$10x - 2x^2$ ✓

c. $10x - 3(2x + 1)$

$10x - 6x - 3$ ✓
 $4x - 3$ ✓

d. $5(x - 1) + 2(3x - 7)$

$5x - 5 + 6x - 14$ ✓
 $= 11x - 19$ ✓

[6 marks]

13. Fully factorise the following expressions

a. $6y + 21$

$3(2y + 7)$ ✓

b. $5x^2 - 15x$

$5x(x - 3)$ ✓

c. $-4p^2 - 2p$

$-2p(2p + 1)$ ✓

d. $3a^2b - 9a^3b^2$

$3a^2b(1 - 3ab)$ ✓

[5 marks]

End of test

